# Mapping Kenya's Electorate

Voter Registration, Presidential Turnout and Demographic Patterns, 2017–2022

***INDEPENDENT RESEARCH REPORT***


## Researcher and author: ARAN FELIX



Study period: 2017–2022 | Coverage: 47 counties, 290 constituencies and 1,450 wards

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Professional portfolio

Research project on GitHub


Aran Felix O.

# Abstract

This study examines changes in Kenya's registered electorate and presidential participation between the 8 August 2017 and 9 August 2022 general elections. The analysis combines Independent Electoral and Boundaries Commission (IEBC) polling-station registration records and presidential-results publications with Kenya National Bureau of Statistics (KNBS) census measures and 2022 county population projections. The validated panel covers 47 counties, 290 constituencies and 1,450 wards. National registration increased from 19,611,423 to 22,120,458 voters, a gain of 2,509,035 (12.79%). However, counted presidential ballots decreased from 15,196,307 to 14,326,751, and turnout declined from 77.49% to 64.77%, a drop of 12.72 percentage points. All 47 counties registered more voters and recorded lower presidential turnout in 2022, although the size of each change varied. Across counties, registration growth was positively associated with turnout change (Pearson's $r = 0.604$; Spearman's $\rho = 0.689$), indicating comparatively smaller declines where registration expanded more rapidly. The analysis also identifies strong population–registration correlations and method-sensitive relationships involving density and selected socioeconomic indicators. These results describe aggregate associations, not individual voting behaviour or causal effects. The study documents source inconsistencies, separates national figures from county-only totals and distinguishes the 2019 census from projected 2022 populations.



**Keywords:** Kenya, voter registration, presidential turnout, electoral geography, counties, census, ecological analysis



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# 1. Introduction

## 1.1 Background and Problem

National election statistics describe overall participation, but they can conceal important differences in the places where citizens register and vote. Registration is an administrative measure of the eligible people recorded on the electoral roll, while turnout reflects the number of ballots counted relative to a defined register. These outcomes are related, but they are not interchangeable. Kenya's 2017 and 2022 general elections provide two comparable points at which to examine whether the growth in registered voters coincided with growth in actual presidential participation. The IEBC publishes detailed registration records and national presidential results, while the KNBS supplies population and socioeconomic indicators that place electoral patterns in geographical context. Using these materials together can reveal contrasts that disappear when each dataset is considered alone. The central challenge is to make those comparisons without confusing reporting levels, election dates, population denominators or explanatory associations. The accompanying research is documented in the professional portfolio case study.

Existing research places these administrative comparisons within the broader context of Kenya's electoral institutions and political competition. Cheeseman et al. (2019) examine the constitutional reforms, political incentives and institutional disputes surrounding the 2017 election, while Cheeseman et al. (2024) assess the changing institutional and political environment of the 2022 contest. Harris (2021) demonstrates that polling-place resource allocation can influence voting costs and turnout in Kenya, showing why registration totals alone cannot measure the ease of casting a ballot. These studies provide plausible contexts for interpreting variation across counties, but they address different questions and employ methods that are not replicated in the present dataset. The analysis therefore treats administrative capacity, mobilization and political institutions as possible avenues for further investigation, not as measured explanations for the observed five-year change.

The underlying records are detailed but heterogeneous. One IEBC file lists registered voters for individual polling-station entries; a different publication reports presidential votes and rejected ballots; KNBS census volumes describe population characteristics in 2019; and a projection document provides estimated county populations for 2022. Each product serves a different purpose. Treating them as a single homogeneous dataset risks creating false turnout measures, misleading population ratios or incorrect geographic maps. The study therefore relies on clear definitions and traceable checks rather than assuming that datasets carrying similar county names have identical universes. The analysis also recognizes that the 2017 presidential election of interest is the election held on 8 August, not the fresh election held on 26 October that year. This distinction is necessary because the two contests are separate electoral events in the IEBC's 2017 publication (IEBC, 2020).

## 1.2 Objectives and Research Questions

The study has four connected objectives. First, it quantifies the change in registered voters and national presidential participation between the 2017 and 2022 general elections. Second, it describes variation in voter registration and turnout across Kenya's 47 counties and the available constituency and ward registration records. Third, it examines whether county registration-growth percentages are associated with changes in county presidential turnout. Fourth, it places the 2022 registration distribution within the context of county population projections and selected socioeconomic characteristics measured by the 2019 census. These objectives correspond directly to the observations contained in the validated evidence package. The research asks where electoral registration expanded

most rapidly, whether presidential participation changed in the same direction, which local units differ from their county totals, and what descriptive patterns are visible when electoral measures are compared with population, density, education and information-technology indicators.

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# 2. Data and Methodology

## 2.1 Sources and Study Design

The study conducts a retrospective, descriptive analysis of official records for two election cycles. The principal registration sources are the IEBC's 2017 and 2022 polling-station voter registers (IEBC, 2017, 2022b). The presidential-results sources are the IEBC's 2020 data report for the 8 August 2017 election and its national tallying-centre declaration dated 15 August 2022 (IEBC, 2020, 2022a). The analysis uses KNBS Volume I of the 2019 Population and Housing Census for observed population geography, Volume IV for selected socioeconomic measures, and the KNBS population-projections summary for 2022 estimates (KNBS, 2019a, 2019b, n.d.). Leadership gazettes and later parliamentary delegation lists form part of the companion dashboard's historical archive, but they do not supply the core registration, presidential-turnout or demographic estimates in this fixed research report.

The unit of analysis depends on the question. The study uses published national presidential totals to calculate the main turnout comparison, geographical county records to examine registration and turnout differences, constituency and ward registration summaries to identify local change, and county-level demographic observations for exploratory associations. The geographical panel contains 47 counties, 290 constituencies and 1,450 wards in each election year. Diaspora and prison registration categories belong in the published national voter-registration totals but are not treated as ordinary counties with physical polygons on the map. The analysis preserves this distinction throughout the paper. The study also retains 2019 population-density and socioeconomic indicators as measurements from 2019, rather than presenting them as newly observed characteristics of either the 2017 or 2022 electorate. The temporal differences prevent direct claims that demographic changes between elections caused participation changes.

## 2.2 Extraction, Geographic Matching and Validation

The source voter registers were distributed as PDFs, and the earlier analysis converted their polling-station entries into standardized records. The validation workflow checked row counts, reported registration totals, geographic codes, duplicated or shifted headers, and consistency across derived aggregation levels. It then produced the consolidated research tables and figures included in evidence release 2017_2022_r01. The preserved package contains 40,883 polling-station records for 2017 and 46,229 for 2022 when national special categories are included. These are record counts and should not automatically be interpreted as numbers of distinct physical buildings. The county-only summaries contain 40,770 and 46,096 such records respectively. Reconciliation across county, constituency and ward summaries produces the same county-only increase of 2,501,030 registered voters. The analysis uses electoral-unit codes and supplementary names to align observations, but a shared code cannot independently certify that all physical boundaries remained unchanged. The reproducible project is documented in the election research repository.

The evidence package maintains separate national and geographic totals because they cover different universes. Published national registered voters totalled 19,611,423 in 2017 and 22,120,458 in 2022. Summing the 47 geographic counties in the station-level datasets gives 19,601,502 and 22,102,532, respectively. The national figures exceed the county-only totals by 9,921 in 2017 and 17,926 in 2022. These are not extraction errors simply because the totals differ: their geographic coverage is different. National registered-voter totals are used only when

presenting national indicators and the county-only registration figures when describing geographic distributions. The 2017 results document also contains a reported inconsistency between its printed national figures and the combined county rows; the final findings retain the published national row while explicitly documenting the problem (IEBC, 2020, p. 19).

### 2.3 Indicator Definitions and Statistical Approach

Registration change is calculated as the 2022 count minus the 2017 count, and percentage growth as that difference divided by the 2017 count, multiplied by 100. Presidential ballots counted equal valid votes plus rejected ballots. The presidential turnout percentage equals counted presidential ballots divided by the registered-voter figure reported in the corresponding presidential-results publication, multiplied by 100. Turnout change is the 2022 percentage minus the 2017 percentage and is expressed in percentage points. For county comparisons, the analysis uses the presidential publication's county denominator rather than substituting the separate polling-station register where the source reports a different count. Constituency and ward records lack a complete matching presidential-vote dataset in this evidence package; consequently, the study does not calculate presidential turnout for those levels. Polling-station changes count source records rather than independently documented station openings or accessibility changes.

The study summarizes electoral differences with counts, rankings, percentages and graphical distributions. To describe the association between county registration-growth percentages and turnout changes, the study reports Pearson's product–moment correlation and Spearman's rank correlation for n = 47 counties. The analysis repeats these measures for county demographic indicators, distinguishing raw-density calculations from log-transformed or rank-based comparisons where those have been documented in the existing outputs. Correlations are exploratory, unadjusted estimates across a census of Kenyan counties, not results of an individual-voter survey or a multivariable causal design. The analysis does not treat p-values as a substitute for substantive interpretation, and the study does not describe a pair of election years as evidence of a persistent time trend. The descriptive rankings are sensitive to baseline county size; therefore, absolute registration gains and proportional growth receive separate treatment.

### 2.4 Population and Socioeconomic Denominators

The KNBS national 2022 population projection is 50,622,914, while the sum of its 47 displayed county projections is 50,622,894, a difference of 20 people. The county sum is preserved when computing geography-based metrics, and cite the printed national projection only when reporting the national estimate (KNBS, n.d., p. 2). Registered voters per 1,000 projected residents equals the county registration count divided by the full projected population, multiplied by 1,000. Because the denominator includes children and other residents who may not be eligible to vote, the result is a contextual ratio rather than an eligible-adult registration rate. The 2019 census indicators for phone ownership and internet use apply to persons aged three years and older; the education variable draws from a different census reporting population (KNBS, 2019a). These definitions limit comparisons of individual rates and require careful language when interpreting correlations.

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# 3. Results

## 3.1 National Registration and Presidential Participation

*Table 1: National registration and presidential participation, 2017 and 2022*

| Indicator | 2017 | 2022 | Change |
|---|---|---|---|
| Registered voters | 19,611,423 | 22,120,458 | +2,509,035 (+12.79%) |
| Valid presidential votes | 15,114,622 | 14,213,137 | −901,485 |
| Rejected presidential ballots | 81,685 | 113,614 | +31,929 |
| All counted presidential ballots | 15,196,307 | 14,326,751 | −869,556 |
| Presidential turnout | 77.49% | 64.77% | −12.72 pp |

Note. National values are taken from the printed IEBC presidential-results totals (IEBC, 2020, p. 19; IEBC, 2022a, p. 3). Percentages and differences were calculated from unrounded counts; pp = percentage points.

Table 1 shows a clear divergence between voter registration and presidential participation. National registration increased by 2,509,035 voters, or 12.79% of the 2017 figure. However, the number of presidential ballots counted fell by 869,556. The reduction reflects fewer valid votes, partly offset by a rise in rejected ballots. When counted ballots are divided by the respective published registered-voter totals, turnout fell from 77.49% to 64.77%, a decline of 12.72 percentage points. The registered-voter base therefore expanded while the counted presidential-vote total contracted. That contrast is important, but it must not be read as proof that additional registrants chose not to vote: the two registers are not person-linked longitudinal datasets, and they cannot identify which individuals entered, left or remained on the roll. The findings instead show how two published system-level indicators changed between elections. The two national comparisons can be inspected directly in Figure 1.

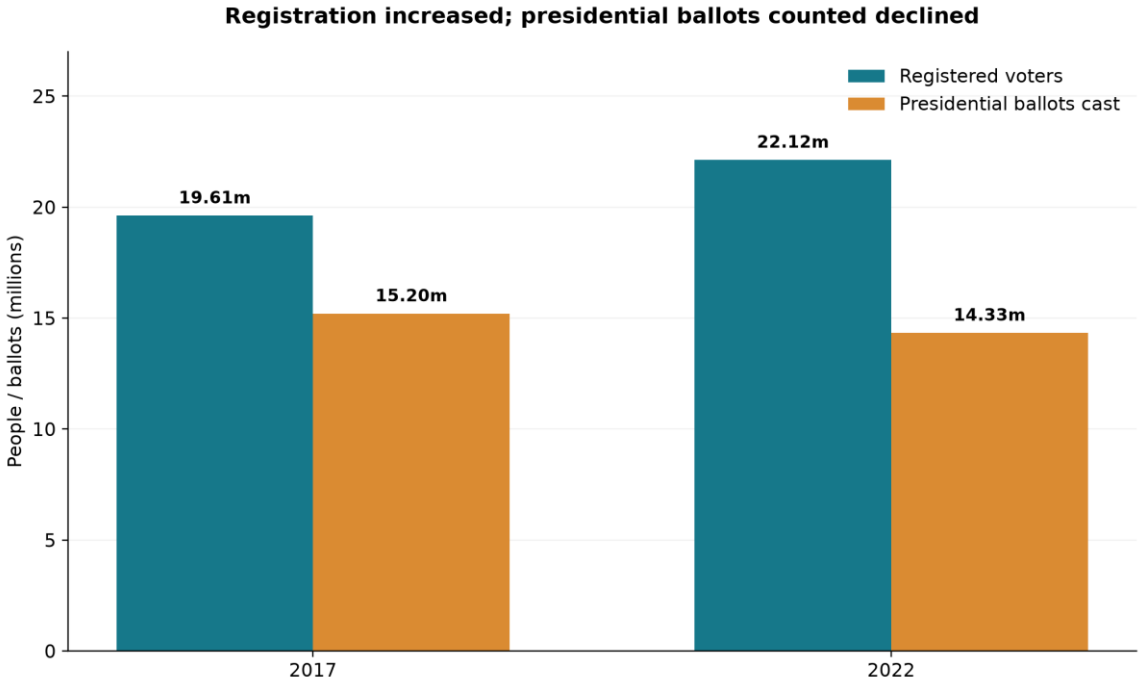



*Figure 1: Registered voters and counted presidential ballots in 2017 and 2022*

Note. Data: IEBC national presidential-results publications. Bars display published totals in millions; counted ballots include valid votes and rejected ballots.

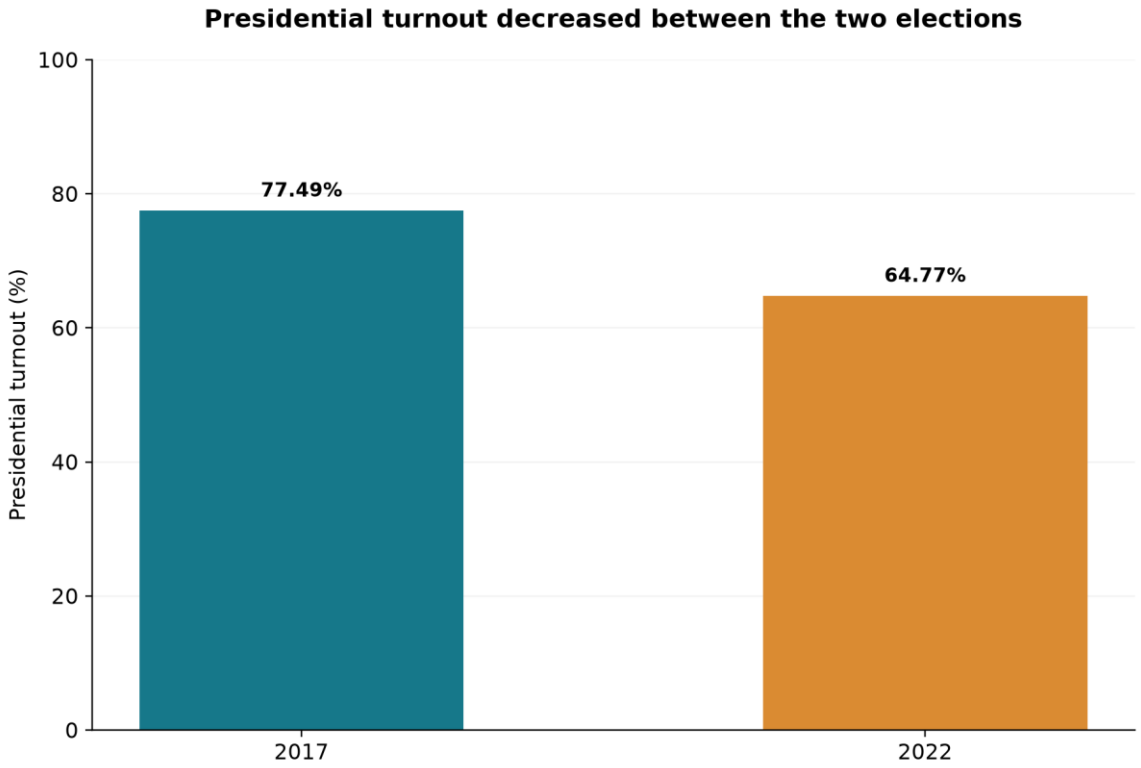



*Figure 2: National presidential turnout across the two general elections*

Note. Turnout = (valid presidential votes + rejected presidential ballots) / published national registered voters × 100. Rounded to two decimals.

### 3.2 Variation Across Counties

All 47 geographical counties recorded higher voter registration in 2022 than in 2017, although the absolute and relative magnitudes varied sharply. Nairobi City recorded the largest increase in people, gaining 164,457 registered voters, followed by Nakuru (105,238) and Kakamega (100,815). In percentage terms, Wajir led with 27.54%, followed by Turkana (24.60%) and Mandera (23.56%). A county with a large register can gain many people while producing a moderate percentage increase, whereas a smaller register can experience substantial proportional growth from a smaller numerical change. The results therefore avoid ranking counties by one measure alone. The sum of county increases was 2,501,030 voters, a county-only change that differs from the published national increase because the latter incorporates special registration categories. The county-only registration series should serve as the denominator for geographic registration maps, not the national total.

*Table 2: Counties with the largest absolute registration gains and percentage growth*

| Largest absolute gains | Voters added | Highest percentage growth | Growth (%) |
|---|---|---|---|
| Nairobi City | 164,457 | Wajir | 27.54 |
| Nakuru | 105,238 | Turkana | 24.60 |
| Kakamega | 100,815 | Mandera | 23.56 |
| Kiambu | 94,088 | Garissa | 23.34 |
| Kisii | 90,430 | West Pokot | 22.08 |

Note. Derived from the matched IEBC county-only registration summaries; ranks use different indicators and should not be conflated.

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Presidential turnout declined in every geographical county between the two elections. Nyandarua registered the largest fall, at 19.50 percentage points, followed by Murang'a (18.79), Kiambu (18.58) and Nyeri (18.53). The unweighted county average decline was approximately 11.32 percentage points, but that arithmetic mean is not the national turnout change because it assigns equal weight to counties with very different electorate sizes. The national result of 12.72 percentage points follows directly from the national ballot and registration totals. When comparing counties, the analysis must also respect source-specific registration denominators: the IEBC's presidential-results county tables may not equal station-register county totals exactly. The county results therefore describe presidential turnout using the presidential publication's recorded voters, while the registration-growth estimates come from the validated station-level series. The county distribution is shown in Figure 3.

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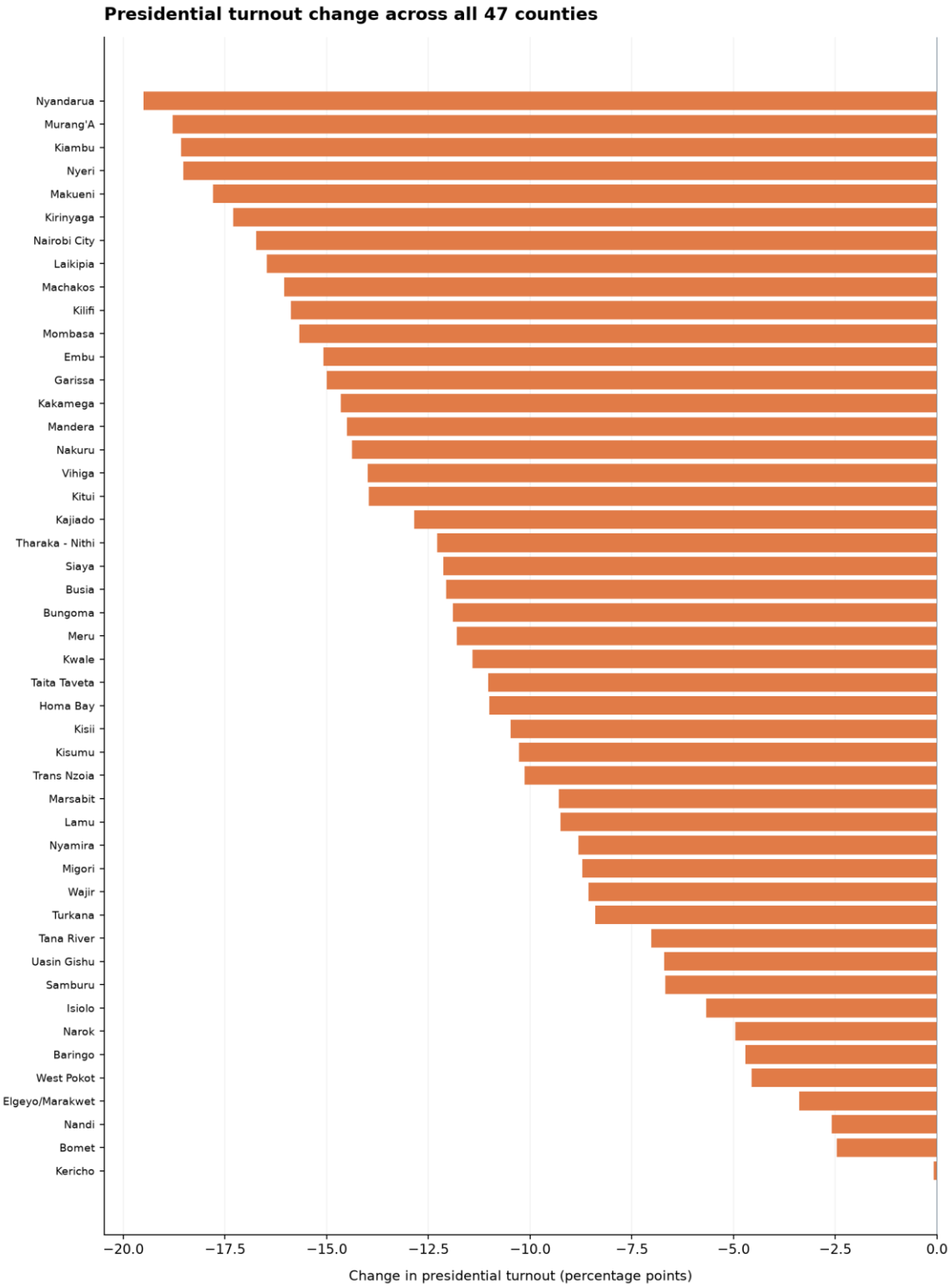



*Figure 3: County-level changes in presidential turnout, 2017–2022*

Note. Negative values indicate turnout declines. Unweighted county comparisons use IEBC county presidential-result denominators; no constituency or ward turnout is estimated.

### 3.3 Constituency and Ward Registration Patterns

The local registration panel provides additional detail beneath the county totals. Of the 290 matched constituencies, 288 registered increases and two recorded decreases. The declining constituencies were Mandera North, down 6,015 registered voters, and Garissa Township, down 1,038. Of the 1,450 matched wards, 1,434 increased while 16 declined. The largest constituency gains occurred in Nyali (20,236), Juja (19,877) and Alego Usonga (18,668).

Local decreases do not contradict universal county registration growth because changes in other constituencies or wards can more than offset a decline within the same county. The county, constituency and ward changes all sum to the same net increase of 2,501,030 voters, demonstrating arithmetic consistency across the geographical aggregation levels. Geographic unit matching is code-based, however, and remains subject to boundary-equivalence and name-standardization limitations.

*Table 3: Constituency registration changes requiring local interpretation*

| Constituency | County | Change, 2017–2022 |
|---|---|---|
| Mandera North | Mandera | −6,015 |
| Garissa Township | Garissa | −1,038 |
| Nyali | Mombasa | +20,236 |
| Juja | Kiambu | +19,877 |
| Alego Usonga | Siaya | +18,668 |

Note. Changes represent registered voters, not turnout. Two negative observations and the three largest absolute gains are shown for contrast; names have been standardized from matched records.

These local patterns make clear why analysis should not stop at county rankings. A decrease within an electoral unit may reflect a substantive registration change, administrative adjustments, record reassignment or other processes that cannot be separated with the available two-year aggregate file alone. Similarly, a higher number of station records in 2022 does not establish that a corresponding number of new physical polling locations became accessible. The observed station-record count increased from 40,883 nationally in 2017 to 46,229 in 2022, but the underlying relationship between locations and polling streams may have changed. The dashboard can still present station-record changes as descriptive administrative indicators, provided its labels explain exactly what was counted. The fixed report does not infer movements of individual registrants between wards or explain local declines without supporting person-level or boundary-change evidence.

### 3.4 Registration Growth and County Turnout Change

Across the 47 counties, the correlation between registration growth (%) and turnout change (percentage points) was positive: Pearson's $r = 0.604$ and Spearman's $\rho = 0.689$. Since turnout changes were negative in every county, this result means that counties with greater registration growth tended, on average, to show less negative changes in presidential participation. It does not mean turnout improved in those counties. The relationship may reflect several overlapping geographic factors, including differences in demographic composition, electoral competition, administration or political context, none of which this pairwise analysis can isolate. The study uses this scatterplot to explain the descriptive relationship rather than turn the fitted line into a prediction or causal estimate. In particular, the study does not establish that registration activity reduced abstention or that a county-level association describes the choices made by particular registered individuals. See the associated scatterplot in Figure 4.

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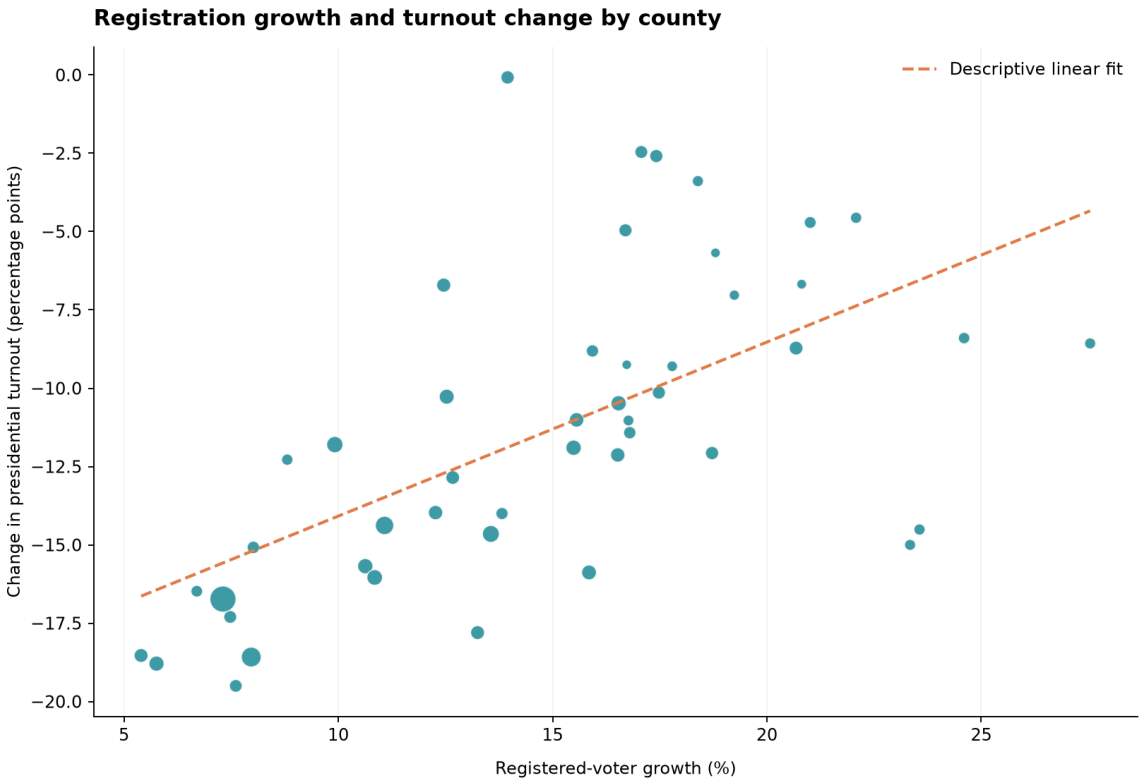



*Figure 4: Association between county registration growth and changes in presidential turnout*

Note. Each point is one county (n = 47); descriptive fitted line only. Pearson's r = 0.604; Spearman's ρ = 0.689. Negative turnout change denotes a decline.

### 3.5 Population Size and Normalized Registration

The 47 counties contained 22,102,532 registered voters in the 2022 station register and an estimated 50,622,894 residents when the KNBS projected county populations were added together. Dividing the register by the projected population yields 436.61 registered voters per 1,000 projected residents. This figure measures the scale of the register relative to the entire estimated resident population; it must not be called the percentage of eligible adults registered. Nyeri had the highest county ratio, approximately 582.52 per 1,000 projected residents, while Garissa had the lowest in the validated series. The strong county population–registration association (Pearson's r = 0.972; Spearman's ρ = 0.911) primarily shows that more populous counties generally contain more registered voters. It does not establish that populous counties performed better at reaching people who were legally eligible to register. The relationship is visualized in Figure 5.

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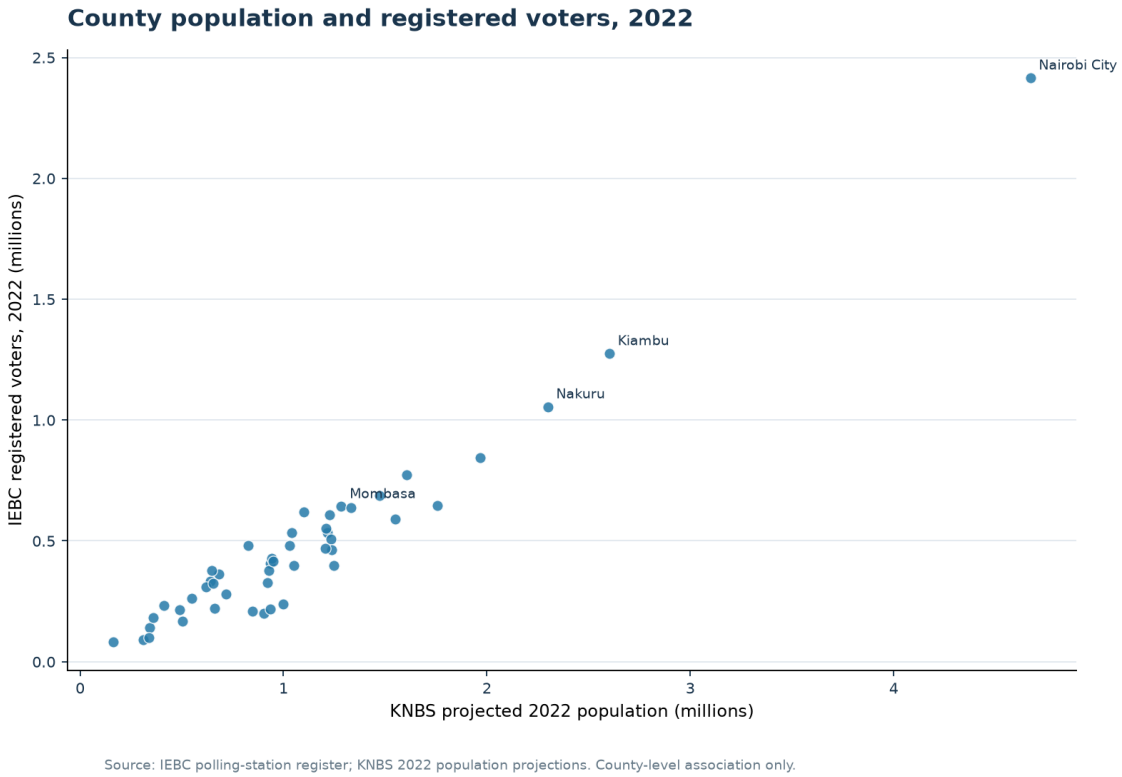



*Figure 5: Projected county population and registered voters in 2022*

Note. Each point represents a county. Population is projected for 2022, and registration is observed in IEBC administrative records; the association is between aggregate counts, not eligible-voter coverage.

### 3.6 Density, Education and Information Technology

The association between 2019 population density and 2022 presidential turnout depends strongly on the statistical measure. Raw-density Pearson correlation was −0.437, while the rank-based Spearman coefficient was −0.010. The nearly zero rank correlation indicates that an apparent negative relationship in the raw-scale Pearson estimate cannot be treated as a stable monotonic county pattern. High-density counties and the distribution of density may influence the raw coefficient, and the evidence also contains alternative log-density specifications. The analysis therefore avoids statements such as "more densely populated counties vote less," which would overstate the data. Density describes geographic population concentration in 2019, not the probability that a particular person voted in 2022. More careful analysis of outliers and additional explanatory variables would be necessary before drawing stronger conclusions about geography and participation.

The 2019 census socioeconomic indicators show exploratory correlations with turnout change from 2017 to 2022. Spearman's ρ was -0.625 for mobile-phone ownership, -0.451 for internet use and -0.421 for the secondary/TVET/university share of the education-table population. Their negative signs indicate that counties with higher measured values of these 2019 indicators tended to have more negative turnout changes, but the correlations neither establish causation nor apply to individuals. The 2019 indicators precede the 2022 election and use population groups that differ from the electoral register. For example, the KNBS's internet-use table covers people aged three years and older rather than eligible voters only (KNBS, 2019a, Table 2.33). The analysis explores multiple unadjusted associations without a preregistered hypothesis or correction for multiple comparisons. These findings should therefore be treated as contextual leads for future research. The comparative coefficient chart appears in Figure 6.

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*Table 4: Selected county-level associations from the validated demographic panel (n = 47)*

| Comparison | Pearson's r | Spearman's ρ |
|---|---|---|
| 2022 projected population vs. registered voters | +0.972 | +0.911 |
| 2019 density vs. 2022 turnout | −0.437 | −0.010 |
| 2019 mobile ownership vs. turnout change | −0.576 | −0.625 |
| 2019 internet use vs. turnout change | −0.445 | −0.451 |
| 2019 education share vs. turnout change | −0.413 | −0.421 |

Note. Values are descriptive county correlations calculated from the consolidated demographic dataset. Source denominators and dates differ, and no causal claim or multiple-testing-adjusted inference is intended.

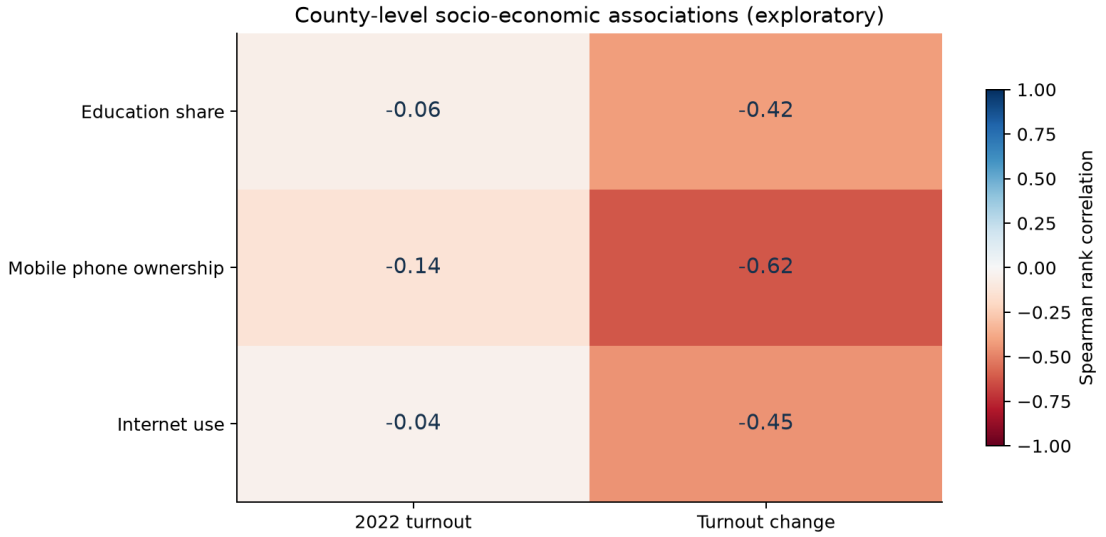



*Figure 6: Exploratory correlations between socioeconomic indicators and presidential turnout*

Note. Spearman correlations across 47 counties. Education and ICT measures come from KNBS 2019 census tables, while presidential outcomes refer to 2017 and/or 2022.

# 4. Discussion

## 4.1 Interpretation of the Central Comparison

The peer-reviewed literature cautions against attributing electoral participation changes to one explanation. Harris (2021) identifies administrative conditions that can influence turnout through the allocation of polling-place resources. Cheeseman et al. (2024) examine Kenya's 2022 election in relation to institutional development and political mobilization, while Cheeseman and Kamencu (2024) document the interaction of urbanization, social change and ethnic political relationships in Central Kenya. These studies demonstrate that local electoral patterns emerge within different institutional and political contexts. However, they do not show that any single mechanism explains the county-level turnout changes calculated here. The present research has no individual-level observations of voter motivations, queues, campaign exposure or trust in electoral institutions. It consequently uses the literature to interpret relevant possibilities while keeping its own findings restricted to documented descriptive differences and ecological associations.

The main result is an electoral participation gap between registration and counted presidential ballots. Registration totals expanded nationally and in every geographic county, yet counted presidential ballots declined at the national level, and turnout declined in all 47 counties. This finding matters for interpreting administrative indicators because increases in registration capacity or register size do not guarantee comparable changes in election-day participation. At the same time, two cross-sectional election records cannot reveal why ballots counted declined. Claims concerning dissatisfaction, political mobilization, administrative barriers, age composition or campaign strategy would require additional sources and more specific research designs. This study therefore treats the contrast as a measured outcome that motivates further investigation rather than as a verdict about voter intentions. Its contribution lies in integrating consistent definitions, validated geography and clear statistical limits into one accessible public record.

## 4.2 What the Geographic Differences Add

The county rankings illustrate why national figures are insufficient. Nairobi City led in absolute increases, while Wajir led in proportional growth. Both statements can be true because the two rankings answer different questions. The universal decline in county turnout also varied in magnitude, with some counties recording small changes and others losing more than 18 percentage points. The local registration files reveal that two constituencies and 16 wards registered decreases even though every county gained voters overall. This nesting demonstrates an aggregation problem: an upward county total can hide opposing movements within the county. It also makes a case for the interactive explorer, where users can move from the headline indicators to validated constituency and ward registration records. The report remains careful not to extend presidential turnout analysis below counties because the necessary vote-count data are not present at those levels.

## 4.3 Reading Demographic Associations Responsibly

Population provides necessary context but is not a substitute for eligibility information. The 2022 county population–registration association is strong largely because counties with more residents tend to have larger electorates. A true comparison of registration coverage would need a consistent estimate of eligible residents and appropriate treatment of residence, citizenship and registration rules. The density analysis shows the importance of methodological sensitivity: Pearson's correlation on raw density and Spearman's rank correlation give very

different pictures of the same turnout outcome. The ICT and education coefficients require further caution because they combine different reference years, denominator populations and unadjusted county-level patterns. These associations could be confounded by settlement patterns, age composition, socioeconomic differences, urbanization or other shared influences. The analysis therefore uses them to generate questions, not to assign electoral behaviour to people who own phones, use the internet or attain a particular education level.

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## 5. Limitations and Evidence Quality

The study has several limitations that affect interpretation. First, the comparison contains only the 2017 and 2022 general elections and cannot establish a continuing electoral trend. Second, the 2017 IEBC report contains documented differences between certain aggregated county values and its printed national presidential totals, including a 34-person register difference and a 30,924-vote difference in valid ballots; these require explicit source-level qualification rather than an undocumented numerical adjustment. Third, national register counts and geographic county-only counts include different reporting categories. Fourth, county and station-level registration totals may not match source-defined presidential-result registration denominators perfectly. Fifth, matching county, constituency and ward codes across years cannot by itself verify unchanged geographic boundaries. Finally, observational correlations across 47 counties do not identify causes, individual behaviours or effects of specific interventions.

Further limitations concern demographic data and documentation. The 2019 census measures refer to an earlier observation year than the 2022 election, while projected populations are modeled estimates rather than census counts. The KNBS projection source prints a national value that exceeds the sum of displayed county values by 20; the difference has been retained, not concealed. Census ICT indicators apply to persons aged three years and older, and the education-table denominator is different. The six selected figures and associated statistical tables retain their original data and disclose their source definitions in the accompanying notes. Historical election winners and later officeholder changes form a separate, approved dashboard feature and should not be included as though they were direct 2017–2022 voter-participation outcomes. The reproducibility records preserve source versions, statistical definitions and documented corrections.

## 6. Conclusion and Next Research Priorities

Kenya’s registered electorate grew by 12.79% between the 2017 and 2022 general elections, while its counted presidential ballots declined by 869,556 and national presidential turnout fell by 12.72 percentage points. Every geographical county added registered voters but also recorded lower presidential turnout. The geographical panel reveals substantial differences in the magnitude of changes and a small number of declining constituencies and wards within counties that grew overall. County correlations describe important contrasts between registration, turnout and demographics, but they cannot identify why any particular person registered or voted. A reproducible, transparent comparison therefore provides more value than an overconfident explanation unsupported by the available data. Future work should examine verified eligible-age denominators, geographic boundary comparability, richer election-context variables and additional election years.

## Research Resources

Visit the professional portfolio and Kenya election research case study

Examine the reproducible election research repository

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Arap Felix O.

## Appendix A. Evidence Provenance and Reproducibility

This report uses the verified evidence release 2017_2022_r01, compiled from the previously validated Phase 5 and Phase 6 outputs. The quantitative tables come from national_indicators_2017_2022.csv, county_rankings.csv, constituency_rankings.csv, ward_rankings.csv, demographic_county_synthesis.csv and demographic_associations_summary.csv. The six displayed figures are copied from the existing inventoried PNGs; the report does not redraw them or modify their original data. Source-page checks focus on the 2017 IEBC report's printed national row (PDF p. 19), the 2022 IEBC national tally declaration (PDF p. 3), the KNBS population-projections table (PDF p. 2) and the KNBS socioeconomic internet-use table (Volume IV, Table 2.33; supplied PDF p. 445). These provide traceable source references but do not establish that every underlying extraction row has been independently re-extracted in the present stage.

Aran Felix O.